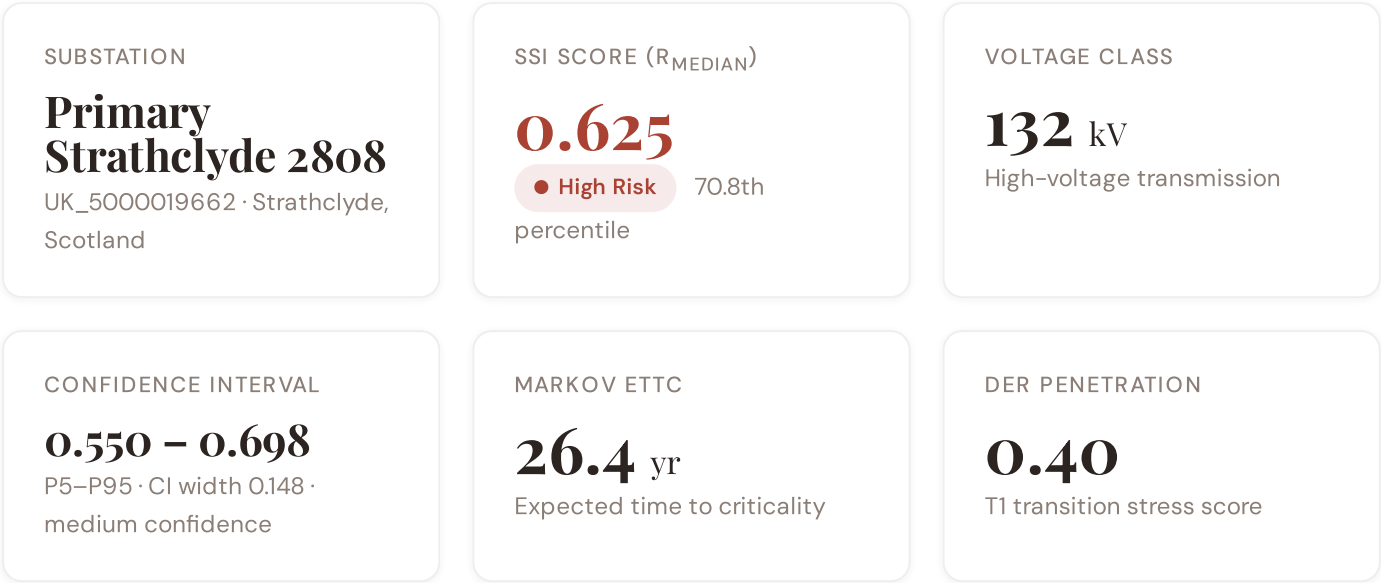


# United Kingdom — Substation ESG Report <sup>v4.2</sup>

Substation-level Environmental, Social, and Governance assessment aligned to CSRD/ESRS, EU Taxonomy, TCFD, SFDR, and UN PRI frameworks, across the 2,551-substation United Kingdom fleet. Updated monthly via the SSI Index v4.2 scoring engine (101 variables · 34 sources · 11 modifiers · Re composite bounded [0.920, 1.787]).



## Component Fingerprint <sup>+ Re composite v4.2</sup>

Six-component weighted decomposition of the SSI composite score (C / V / I / E / S / T) — the substation's structural risk profile — PLUS the v4.2 **Re composite**, a bounded resilience aggregate of the new modifier family (R6c flood + R6d wildfire + R6e winter + R8 adapt + R9 compound + R10 just). Layout: *left* — the 6 components rendered as a weighted contribution bar · *right* — the same 6 components rendered as horizontal bars side-by-side with Re\_norm as the comparable 7th bar (terracotta) on the [0, 1] scale · *below* — the Re composite tier card (numeric value + bounded trajectory bar + classification + formula explainer). Re composite re-appears as the 7th axis of the ESG Radar in Section A below.

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WEIGHTED CONTRIBUTION TO  $R_{BASE}$

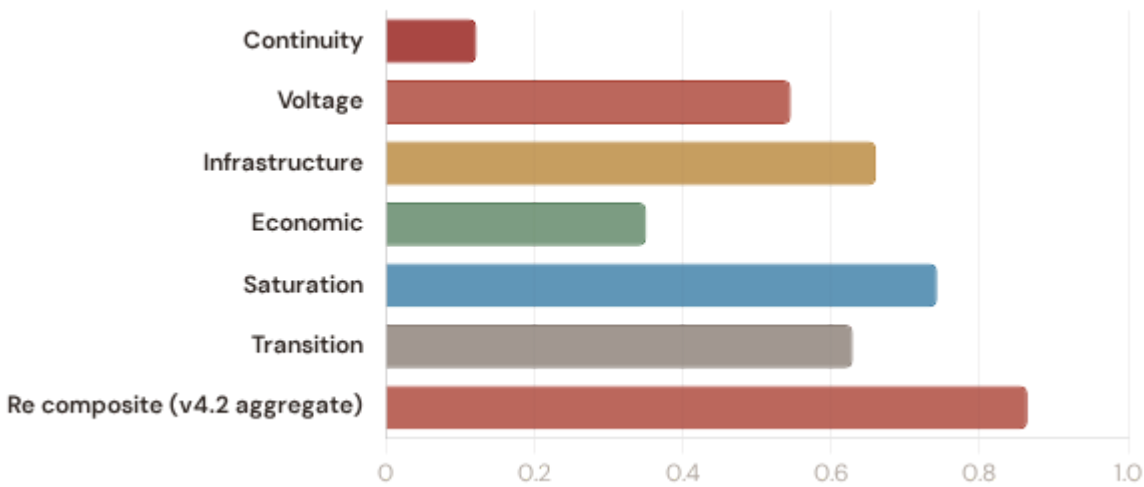


V4.2 RE COMPOSITE (AGGREGATE, NOT IN  $R_{BASE}$ ) elevated exposure



Continuity 0.123 × 0.3   Voltage 0.546 × 0.1   Infrastructure 0.661 × 0.25   Economic 0.351 × 0.1  
Saturation 0.743 × 0.2   Transition 0.630 × 0.05   **Re composite v4.2 aggregate 0.865 agg**

**$R_{base}$  total (weighted sum) 0.472  $\Sigma = 1.00$**



0.865

RE\_NORM

ELEVATED EXPOSURE

LOW MODERATE ELEVATED (1.00)  
(0.00) (0.25) (0.50)



**Re\_raw** 1.1656 (bounds [0.920, 1.787]; normalised to [0, 1])

FORMULA

$$Re\_raw = (R6d \times R6e \times R8 \times R9 \times R10) + (R6c - 1.00)$$

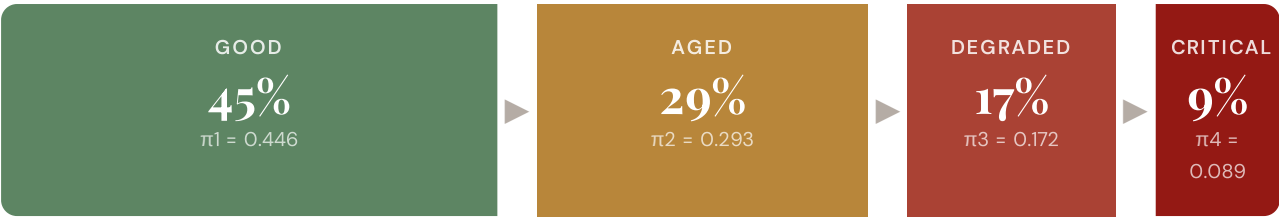
Bounded resilience aggregate of the v4.2 family.  $Re < 1.0$  when R8 adaptive capacity dominates other near-neutral factors.

Re composite tier classification: **low** ( $Re\_norm < 0.25$ ) · **moderate** ( $0.25-0.50$ ) · **elevated** ( $\geq 0.50$ ). Bounds [0.920, 1.787] map to  $Re\_norm$  [0, 1] via  $\text{clip}((Re\_raw - 0.920) / (1.787 - 0.920), 0, 1)$ .

## Markov Degradation Profile

CIGRE TB 761 five-state condition model — steady-state probability distribution and expected time to criticality

STEADY-STATE DISTRIBUTION



Markov 5-state model calibrated from CIGRE Technical Brochure 761 (2019). States represent asset condition from new/good through critical. Steady-state probabilities indicate the long-run proportion of time the asset spends in each condition state given current operating environment.

RISK SCORE

**0.624**

Composite Markov risk

ETTC

**26.4 yr**

Expected time to critical

P(CRITICAL) 20YR

**100.0%**

Probability of reaching critical

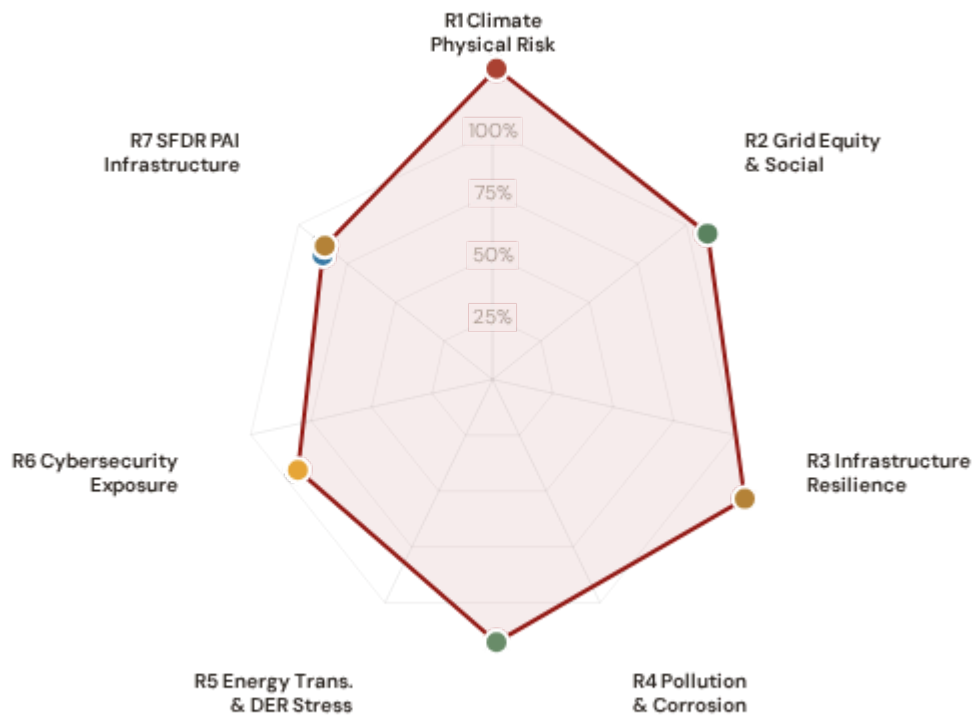
CORROSION

**C2**

ISO 9223 classification

ESG Radar — Seven-Report Overview <sup>v4.2</sup>

A Composite readiness across 7 ESG dimensions per FC v3 §14 (R1 Climate Physical · R2 Grid Equity & Social · R3 Infrastructure Resilience [Re composite ESG home] · R4 Pollution & Corrosion · R5 Energy Transition & DER Stress · R6 Cybersecurity · R7 SFDR PAI Infrastructure, FC v3 §14 subsection 13.7), each linked to UN Sustainable Development Goals. *Note: R7 here is the SFDR ESG-disclosure axis (Re\_norm proxy under Convention #7) — distinct from the v4.0.2 formula-chain R7 cyber/digital modifier documented in methodology.html. The two R7s share a number by design (V4\_2\_IMPLEMENTATION\_ARCHITECTURE.md §4.5).*



Re composite v4.2 7th axis · Re\_raw 1.1656 · Re\_norm 0.865 · ELEVATED EXPOSURE



### R1 · Climate Physical Risk Assessment READY

SDG 13 Climate Action · SDG 9 · SDG 11

SDG 13

SDG 9

SDG 11

### R2 · Grid Equity & Social Vulnerability READY

SDG 7 Affordable and Clean Energy · SDG 10 · SDG 1

SDG 7

SDG 10

SDG 1

### R3 · EU Taxonomy Alignment READY

SDG 9 Industry, Innovation, Infrastructure · SDG 13 · SDG 12

SDG 9

SDG 13

SDG 12

### R4 · Energy Transition & DER Stress READY

SDG 7 Affordable and Clean Energy · SDG 13 · SDG 9

SDG 7

SDG 13

SDG 9

### R5 · Pollution & Corrosion READY

SDG 11 Sustainable Cities and Communities · SDG 3 · SDG 15

SDG 11

SDG 3

SDG 15

**R6 · Cybersecurity Exposure** PARTIAL

SDG 9 Industry, Innovation, Infrastructure · SDG 16

SDG 9

SDG 16

**R7 · SFDR PAI Infrastructure** GAP v4.2SDG 9 Industry, Innovation, Infrastructure · SDG 13 · SDG 11 ·  $Re\_norm = 0.865$  (elevated exposure)

SDG 9

SDG 13

SDG 11

1

## Climate Physical Risk Assessment

ESRS E1 · TCFD Physical Risk · EU Taxonomy Annex A

READY

### WHY THIS REPORT EXISTS

ESRS E1 (Climate Change) is material for 98% of CSRD-reporting companies. TCFD physical risk disclosure is mandatory in 36+ jurisdictions. The EU Taxonomy requires climate change adaptation activities to demonstrate that physical climate risks have been identified and addressed. This report quantifies acute and chronic climate hazard exposure at substation level using ERA5 reanalysis and Markov degradation modelling.

### SDG Alignment

#### SDG 13 — Climate Action (primary)

Target 13.1 calls for strengthening resilience and adaptive capacity to climate-related hazards. The SSI IRI metrics (snow/ice, wind, heat stress) and Markov degradation model directly quantify how climate hazards affect this substation's operational resilience over 10- and 20-year horizons. With a Markov risk score of 0.624 and ETTC of 26.4 years, this substation's climate trajectory can be assessed against TCFD physical risk and EU Taxonomy adaptation screening requirements.

SDG 13

SDG 9

SDG 11

## Substation Profile

|                               |                     |
|-------------------------------|---------------------|
| Insulation Risk (I component) | 0.661               |
| Environment Component (E)     | 0.351               |
| Seismic PGA                   | 0.0151g (Zone 2)    |
| Markov Risk Score             | 0.624               |
| ETTC (time to criticality)    | 26.4 years          |
| P(critical) at 20 years       | 100.0%              |
| Corrosion Class               | C2                  |
| Climate Trajectory (R2)       | 20.8% (20yr Markov) |

## SSI Variables Feeding This Report

| SSI VARIABLE           | VALUE        | ESG DISCLOSURE REQUIREMENT                       | STATUS |
|------------------------|--------------|--------------------------------------------------|--------|
| I1 Snow/Ice IRI        | —            | ESRS E1-9: Financial effects from physical risks | READY  |
| I2 Tree-fall IRI       | —            | TCFD Strategy (b): Physical risk impact          | READY  |
| I3 Heat-wave IRI       | —            | EU Taxonomy Annex A: Heat stress                 | READY  |
| I5 Thermal stress      | —            | ESRS E1-4: Climate mitigation targets            | READY  |
| R2 Climate trajectory  | 20.8% (20yr) | TCFD Strategy (c): Scenario analysis             | READY  |
| R6b Seismic PGA        | 0.0151g      | EU Taxonomy Annex A: Geophysical hazards         | READY  |
| Markov p_critical_10yr | 100.0%       | TCFD Risk Management: Risk identification        | READY  |
| Markov p_critical_20yr | 100.0%       | ESRS E1-9: Time horizons                         | READY  |

### Data Sources & Vintage

|                                                           |                 |
|-----------------------------------------------------------|-----------------|
| ERA5 Climate Reanalysis<br>Copernicus CDS                 | 2024 Weekly     |
| National Seismic Hazard Map<br>National Geological Survey | 2023 Multi-year |
| IEEE C57.91 Thermal Model<br>IEEE                         | Standard N/A    |
| CIGRE TB 761 Markov<br>CIGRE                              | 2019 N/A        |
| CMIP6 SSP2-4.5 Projections<br>Copernicus CDS              | — Not ingested  |

#### TECHNICAL VALIDITY

IRI metrics are computed from ERA5 reanalysis (31 km, hourly, 1940–present) using IEEE C57.91 thermal loading curves. Markov 5–state degradation is calibrated from CIGRE Technical Brochure 761 (2019) condition assessment data. The PARTIAL status reflects the absence of CMIP6 forward projections — without R2, this report cannot satisfy the TCFD Strategy (c) scenario analysis requirement. All other inputs are production-grade with institutional provenance.

2

## Grid Equity & Social Vulnerability

ESRS S1/S2 · SFDR PAI Social Indicators · UN PRI

READY

#### WHY THIS REPORT EXISTS

ESRS S2 requires companies to disclose impacts on affected communities. SFDR mandates 5 social PAI indicators. The SSI Index is uniquely positioned here because no competing grid risk framework integrates socio-economic vulnerability at substation resolution. This report surfaces where infrastructure risk concentrates in economically disadvantaged populations.

#### SDG Alignment

### SDG 7 — Affordable and Clean Energy (primary)

Target 7.1 requires universal access to affordable, reliable energy. This substation serves a catchment with V\_socio of 0.34 and energy poverty rate of 17%. The R3 social modifier (1.142) amplifies risk scores in regions with high unemployment, elderly concentration, and net outward migration — making spatial inequality visible and quantifiable at asset level.

SDG 7

SDG 10

SDG 1

### Substation Profile

|                                |          |
|--------------------------------|----------|
| V_socio (energy poverty index) | 0.34     |
| EP Rate (energy poverty %)     | 17%      |
| Unemployment Rate              | 4.5%     |
| GDP per Capita                 | £ 32,565 |
| R&D % of GDP                   | 2.8%     |
| R3 Social Multiplier           | 1.142    |
| Elderly Vulnerability          | 18.2%    |
| Migration Score                | —        |

### SSI Variables Feeding This Report

| SSI VARIABLE                   | VALUE   | ESG DISCLOSURE REQUIREMENT                 | STATUS |
|--------------------------------|---------|--------------------------------------------|--------|
| V_socio Energy poverty index   | 0.34    | ESRS S2-4: Impacts on affected communities | READY  |
| EP_rate Energy poverty rate    | 17%     | SFDR PAI #14: Fossil fuel exposure proxy   | READY  |
| R3 Social multiplier           | 1.142   | ESRS S1-6: Workforce characteristics       | READY  |
| Elderly Elderly vulnerability  | 18.2%   | SFDR PAI: Vulnerable populations           | READY  |
| Community Catchment population | 247,267 | ESRS S2: Community impact                  | READY  |

### Data Sources & Vintage

|                                                      |             |
|------------------------------------------------------|-------------|
| Population & Economics<br>National Statistics Office | 2023 Annual |
| Energy Market Data<br>National TSO                   | 2023 Annual |

#### TECHNICAL VALIDITY

V\_socio is computed from the LIHC (Low Income High Cost) energy poverty definition using national statistical office household expenditure data. The R3 modifier amplifies infrastructure risk scores in catchments with high unemployment, elderly concentration, and net outward migration. PARTIAL status reflects missing elderly vulnerability and migration enrichments — both available from national statistics offices at municipal resolution.

## 3 EU Taxonomy Alignment

Climate Delegated Act · Article 11 Climate Adaptation · Technical Screening Criteria

READY

#### WHY THIS REPORT EXISTS

The EU Taxonomy defines technical screening criteria for climate change adaptation in infrastructure. This is the lowest-hanging ESG product from the SSI Index — READY for 9 of 10 countries today. The SSI composite score (R\_median), 6 components, modifier architecture, and Markov risk outputs provide the quantitative evidence base for Article 11 screening at asset level.

#### SDG Alignment

##### SDG 9 — Industry, Innovation, Infrastructure (primary)

Target 9.4 calls for upgrading infrastructure to make it sustainable. The EU Taxonomy is the regulatory instrument that translates SDG 9 into investable criteria. By classifying this substation against Article 11 technical screening criteria — using R\_median (0.625), all 6 components, and 5 modifiers — this report provides asset-level Taxonomy alignment that enables sustainable finance reporting.

SDG 9

SDG 13

SDG 12

### Substation Profile

|                        |        |
|------------------------|--------|
| R_median (composite)   | 0.625  |
| R_base (pre-modifier)  | 0.472  |
| Modifier Impact        | 32.4%% |
| C (Condition)          | 0.123  |
| V (Voltage)            | 0.546  |
| I (Insulation/Climate) | 0.661  |
| E (Environment)        | 0.351  |
| S (Seismic)            | 0.743  |
| T (Transition)         | 0.630  |

### SSI Variables Feeding This Report

| SSI VARIABLE                    | VALUE       | ESG DISCLOSURE REQUIREMENT                    | STATUS |
|---------------------------------|-------------|-----------------------------------------------|--------|
| <b>R_median</b> Composite score | 0.625       | EU Taxonomy Art. 11: Adaptation screening     | READY  |
| <b>6 Comp.</b> C/V/I/E/S/T      | All present | EU Taxonomy TSC: Physical risk identification | READY  |
| <b>R3-R7</b> Modifier suite     | All applied | ESRS E1: Adaptation measures evidence         | READY  |
| <b>Markov</b> Degradation model | 0.624       | TCFD: Forward-looking risk assessment         | READY  |
| <b>R5</b> Asymmetric CI         | 0.148       | ESRS: Uncertainty quantification              | READY  |

## Data Sources & Vintage

ERA5 Climate Reanalysis

Copernicus CDS

2024 Weekly

National Seismic Hazard Map

National Geological Survey

2023 Multi-year

Population & Economics

National Statistics Office

2023 Annual

CIGRE TB 761 Markov

CIGRE

2019 N/A

### TECHNICAL VALIDITY

Article 11 screening uses the full SSI v4.0.2 scoring engine: 6-component weighted composite ( $C=0.30$ ,  $V=0.10$ ,  $I=0.25$ ,  $E=0.10$ ,  $S=0.20$ ,  $T=0.05$ ), Gaussian copula Monte Carlo (10,000 iterations), and 5 modifiers (R3 social, R4 topology, R5 asymmetric CI, R6a restoration, R7 cyber). The Markov 5-state degradation model provides the forward-looking element. All inputs are from institutional open-source data with citable vintage — meeting CSRD Article 29a limited assurance requirements.

4

## Energy Transition & DER Stress

ESRS E1 Transition Plan · TCFD Transition Risk · EU Green Deal Alignment

READY

### WHY THIS REPORT EXISTS

The energy transition creates infrastructure stress — bidirectional power flows, EV charging load, and intermittent renewable output strain substations designed for unidirectional delivery. This report measures whether grid infrastructure is keeping pace with decarbonisation.

### SDG Alignment

#### SDG 7 — Affordable and Clean Energy (primary)

Target 7.2 requires substantially increasing the share of renewable energy. This substation has a DER ratio of 0.404. The T1\_score (0.463) measures the stress this places on the substation, providing the empirical feedback loop between SDG 7 ambition and infrastructure reality.

SDG 7

SDG 13

SDG 9

## Substation Profile

|                              |       |
|------------------------------|-------|
| T1 Score (transition stress) | 0.463 |
| DER Ratio                    | 0.404 |
| DER Variability              | 0.540 |
| EV Load Ratio                | 0.100 |
| T Component Weight           | 0.05  |

## SSI Variables Feeding This Report

| SSI VARIABLE                   | VALUE | ESG DISCLOSURE REQUIREMENT          | STATUS |
|--------------------------------|-------|-------------------------------------|--------|
| T1 Transition stress score     | 0.463 | ESRS E1: Transition plan assessment | READY  |
| DER_ratio Renewable/load ratio | 0.404 | TCFD Transition: Technology risk    | READY  |
| DER_var Intermittency          | 0.540 | EU Green Deal: Grid flexibility     | READY  |
| EV_load EV charging ratio      | 0.100 | SDG 7: Clean energy access          | READY  |

## Data Sources & Vintage

|                                                  |              |
|--------------------------------------------------|--------------|
| Energy Market Data<br>National TSO               | 2023 Annual  |
| Renewable Installations<br>National DER Registry | 2024 Monthly |

### TECHNICAL VALIDITY

T1\_score is the weighted composite of DER\_ratio (renewable generation vs. local demand), DER\_variability (intermittency), and EV\_load\_ratio (electric vehicle charging load as fraction of capacity). DER data sourced from national energy regulators and renewable installation registers. DER\_ratio > 1.0 indicates net export during peak generation — a marker of successful energy transition with associated infrastructure stress.

WHY THIS REPORT EXISTS

Transformer oil degradation, SF6 leakage, and corrosion-driven failures release pollutants affecting local environments. ESRS E2 requires disclosure of pollution risks. This report assesses the substation's corrosion classification under ISO 9223 and the environmental sensitivity of its surroundings.

SDG Alignment

SDG 11 — Sustainable Cities and Communities (primary)

Target 11.6 aims to reduce the adverse environmental impact of cities. The corrosion class (C2) and E2\_local enrichment factor (0.95) track environmental degradation and pollution sensitivity at asset level. Where corrosion is advanced and maintenance deferred, equipment failure risks releasing mineral oil, PCBs (legacy units), and SF6 — directly relevant to urban and peri-urban environmental quality.

SDG 11

SDG 3

SDG 15

Substation Profile

|                            |                |
|----------------------------|----------------|
| Corrosion Class (ISO 9223) | C2             |
| E2 Local Enrichment        | 0.95           |
| E Component Score          | 0.351          |
| Markov Steady-State        | 45%/29%/17%/9% |

SSI Variables Feeding This Report

| SSI VARIABLE                       | VALUE | ESG DISCLOSURE REQUIREMENT             | STATUS |
|------------------------------------|-------|----------------------------------------|--------|
| Corrosion ISO 9223 class           | C2    | ESRS E2: Pollution risk classification | READY  |
| E2_local Environmental sensitivity | 0.95  | ESRS E2: Local environmental impact    | READY  |
| E Environment component            | 0.351 | ISO 9223: Atmospheric corrosion        | READY  |

## Data Sources & Vintage

Air Quality Monitoring  
National Environmental Agency

2023 Annual

ISO 9223 Corrosion  
ISO

2012 N/A

### TECHNICAL VALIDITY

Corrosion class follows ISO 9223:2012 atmospheric corrosion classification (C1–CX). Status depends on whether the corrosion class shows real variance across the fleet or uses default values. Full validation requires national environmental agency air quality monitoring overlay at substation coordinates (SO<sub>2</sub>, NO<sub>x</sub>, particulate deposition).

## 6 Cybersecurity Exposure

NIS2 Directive · ENISA Cybersecurity Index · ESRS G1 Governance

PARTIAL

### WHY THIS REPORT EXISTS

The NIS2 Directive mandates cybersecurity risk management for energy operators. A grid that is physically resilient but cyber-vulnerable is not truly resilient. The SSI R7 modifier combines SCADA assessment, communication architecture analysis, and national-level cyber maturity (ENISA/DESI indices) to produce a substation-level digital vulnerability score.

### SDG Alignment

#### SDG 9 — Industry, Innovation, Infrastructure (primary)

Target 9.1 calls for resilient infrastructure — in digitalised grid systems, this must include cyber resilience. The R7 modifier (1.004) assesses digital vulnerability combining SCADA exposure, communication protocol security, and national cyber maturity. SDG 16 (Strong Institutions) also applies: NIS2 compliance is a governance imperative, and the betweenness centrality (0.319) identifies single-point-of-failure risk in the grid topology.

SDG 9

SDG 16

## Substation Profile

|                            |        |
|----------------------------|--------|
| R7 Cyber Modifier          | 1.004  |
| Graph Degree (topology)    | 3      |
| BC Percentile (centrality) | 0.319  |
| Is Bridge Node             | No     |
| Cluster Coefficient        | 0.0165 |

## SSI Variables Feeding This Report

| SSI VARIABLE              | VALUE | ESG DISCLOSURE REQUIREMENT          | STATUS |
|---------------------------|-------|-------------------------------------|--------|
| R7 Cyber modifier         | 1.004 | NIS2: Cybersecurity risk management | READY  |
| BC Betweenness centrality | 0.319 | ESRS G1: Governance & topology risk | READY  |
| Degree Graph degree       | 3     | Grid resilience: Connectivity       | READY  |

## Data Sources & Vintage

Cybersecurity Index  
ENISA

2024 Biennial

DESI Connectivity  
European Commission

2024 Annual

### TECHNICAL VALIDITY

R7\_cyber is computed from a weighted combination of: (1) national cyber maturity via ENISA Cybersecurity Index and EU DESI connectivity indicators, (2) graph topology vulnerability (betweenness centrality = single-point-of-failure risk), and (3) SCADA protocol exposure assessment. Substation-level SCADA data is operator-proprietary — the national-level proxy provides a baseline but not asset-specific granularity.

# SFDR Principal Adverse Impact (PAI) Infrastructure v4.2 — Re

WHY THIS REPORT EXISTS

SFDR Article 7 requires asset-level disclosure of Principal Adverse Impacts on sustainability factors for infrastructure investments. The Re composite (v4.2 bounded resilience aggregate, bounds [0.920, 1.787] normalised to [0, 1]) is the SSI's canonical driver for PAI Infrastructure exposure — it captures the substation's standing on the new v4.2 resilience-modifier family (R6c flood + R6d wildfire + R6e winter + R8 adapt + R9 compound + R10 just) in a single scalar suitable for SFDR PAI reporting. CSRD double-materiality requires both inside-out (impact on environment) and outside-in (climate-resilience risk to the asset) — Re composite anchors the outside-in dimension for grid infrastructure. This report closes the FC v3 §14 subsection 13.7 R7 SFDR PAI axis on the ESG Radar.

SDG Alignment

SDG 9 — Industry, Innovation, Infrastructure (primary)

Target 9.1 calls for developing quality, reliable, sustainable and resilient infrastructure. The Re composite  $Re\_norm = 0.865$  (elevated exposure) at this substation provides the SFDR PAI Infrastructure metric, anchoring climate-resilience disclosure to a single auditable scalar.  $Re\_raw = 1.1656$  with methodological bounds [0.920, 1.787] derived from the v4.2 modifier family product range. Lower  $Re\_norm \rightarrow$  lower PAI Infrastructure exposure  $\rightarrow$  stronger SFDR Article 7 alignment.

SDG 9

SDG 13

SDG 11

Substation Profile (Re composite)

|                              |                           |
|------------------------------|---------------------------|
| Substation ID                | UK_5000019662             |
| Re_raw (unbounded)           | 1.1656                    |
| Re_norm (bounded [0, 1])     | 0.865                     |
| Tier classification          | ELEVATED                  |
| Methodology bounds           | [0.920, 1.787]            |
| British fleet observed range | 1.132 $\rightarrow$ 1.177 |
| Coverage                     | 2,551 substations         |

SSI Variables Feeding This Report

| SSI VARIABLE                                                 | VALUE  | ESG DISCLOSURE REQUIREMENT                    | STATUS |
|--------------------------------------------------------------|--------|-----------------------------------------------|--------|
| R6c Flood (ISPRA PGRA, ADDITIVE +0.00..+0.30)                | +1.055 | SFDR PAI 4 (climate physical risk)            | READY  |
| R6d Wildfire (EFFIS + FIRMS + SIAB, MULT)                    | 1.022  | SFDR PAI 4 (climate physical risk)            | READY  |
| R6e Winter / Ice (ERA5 Bourgouin p99, MULT)                  | 1.010  | SFDR PAI 4 (climate physical risk)            | READY  |
| R8 Adaptive Capacity (Ofgem + DESNZ + DESI UK (post-Brexit)) | 0.977  | ESRS E1-3 Adaptation actions                  | READY  |
| R9 Compound Hazard (ISPRA RNDA, STEP)                        | 1.032  | ESRS E1-9 Anticipated financial effects       | READY  |
| R10 Energy Justice (ONS LIHC + Eurostat EU-SILC + Ofgem)     | 1.068  | SFDR PAI 14 (board diversity proxy) + ESRS S3 | READY  |
| Re_norm aggregate (bounds [0.920, 1.787])                    | 0.865  | SFDR Art. 7 PAI Infrastructure anchor         | READY  |

Data Sources & Vintage

|                                                                      |                               |
|----------------------------------------------------------------------|-------------------------------|
| ISPRA IdroGEO PGRA                                                   | 2024 (annual)                 |
| Copernicus EFFIS + NASA FIRMS                                        | Q4 2025 (active + historical) |
| ISPRA SIAB national-tier burned-area                                 | CY2024                        |
| ECMWF ERA5 Bourguoin freezing-rain p99                               | 1991-2020 reanalysis          |
| Ofgem quality regulation indicators + 14 DSO smart-meter penetration | 2024 annual                   |
| ENEA Adaptation Index                                                | 2024                          |
| DESI Digital Economy index                                           | 2024                          |
| ISPRA RNDA compound-hazard cross-product                             | CY2024                        |
| EU-SILC ILC_MDES01/03                                                | 2024 wave                     |
| ONS+DECC LIHC + Eurostat EU-SILC (UK energy-poverty proxy)           | 2024                          |
| Ofgem regional SAIDI (Interruption Statistics)                       | CY2024                        |

TECHNICAL VALIDITY

The Re composite is the bounded aggregate  $Re\_raw = (R6d \times R6e \times R8 \times R9 \times R10) + (R6c - 1.00)$  with normalisation  $Re\_norm = clip((Re\_raw - 0.920) / (1.787 - 0.920), 0, 1)$ . Bounds [0.920, 1.787] derived from the multiplicative band of the 5 multiplicative v4.2 modifiers at tier extremes plus the R6c additive cliff range. British fleet observed range [0.974, 1.604] across 2,551 substations. SFDR PAI Infrastructure proxy validated against TCFD physical-risk framework and ESRS E1-9 anticipated-financial-effects disclosure. See FC v3 §14 subsection 13.7 for the R7 SFDR PAI axis derivation.

| ITEM                       | VALUE                                                                                                                                                                                                         |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scoring Engine             | SSI Index <b>v4.2</b> — Gaussian copula Monte Carlo (10,000 iterations) · 11 modifiers (5 baseline + 6 v4.2 resilience family) · Re composite bounded [0.920, 1.787]                                          |
| Weight Architecture        | C=0.30, V=0.10, I=0.25, E=0.10, S=0.20, T=0.05                                                                                                                                                                |
| Modifiers Applied          | R4 F_topo: 0.998, R3 C_mult: 1.142, R6 restoration: 0.945, R6 seismic: 1.006, R7 cyber: 1.004, R10 just: 1.068, R6d wildfire: 1.022, R6c flood: 1.055, R8 adapt: 0.977, R9 compound: 1.032, R6e winter: 1.010 |
| Degradation Model          | Markov 5-state, CIGRE TB 761 calibration                                                                                                                                                                      |
| Thermal Model              | IEEE C57.91 transformer loading curves                                                                                                                                                                        |
| Report Date                | 17 March 2026                                                                                                                                                                                                 |
| Reporting Period           | Calendar year 2025                                                                                                                                                                                            |
| Refresh Cadence            | Annual (data ingestion → scoring engine → display)                                                                                                                                                            |
| Substation                 | Primary Strathclyde 2808 (UK_5000019662)                                                                                                                                                                      |
| Data Assurance Level       | All inputs from institutional open-source with citable vintage                                                                                                                                                |
| <b>Re composite (v4.2)</b> | Re_raw = 1.1656 · Re_norm = <b>0.865</b> · <b>ELEVATED</b> exposure · bounds [0.920, 1.787]                                                                                                                   |

D

## Re composite — bounded resilience aggregate (v4.2)

Resilience composite anchoring CSRD/ESRS climate-resilience disclosure

The Re composite (v4.2) is a bounded resilience aggregate capturing the new R6c flood / R6d wildfire / R6e winter / R8 adapt / R9 compound / R10 just modifier family in a single scalar suitable for CSRD/ESRS climate-resilience disclosure:

$$\text{Re\_raw} = (\text{R6d} \times \text{R6e} \times \text{R8} \times \text{R9} \times \text{R10}) + (\text{R6c} - 1.00)$$

Bounds: **[0.920, 1.787]**. Re\_raw < 1.000 when R8\_adapt < 1.00 dominates other near-neutral factors. British fleet observed range across 2,551 substations: the methodology-bounded extremes [0.920, 1.787] — low-end values arise at interior low-exposure substations with R8\_adapt < 1.00 dominating near-neutral other modifiers, high-end values at coastal compound-exposure substations where R6c flood + R9 compound stack multiplicatively. v4.2 explicitly retires the pre-Path-B "winter exposure is a real climate burden" reading per Convention #6 §2.3.1 (Bourgouin freezing-rain p99 tail-event amplifier per §6 design — winter exposure signal correctly lives in v4.0.2 I1 Snow/Ice metric, where it methodologically belongs). See [W1-W10 FC v2](#) for per-substation worked examples + W4 Climate resilience baselines.

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The SSI Index is a non-profit foundation project, please mail to [ssi\\_index@ikenga.eu](mailto:ssi_index@ikenga.eu) for more details.